An End-to-End Performance Analytics Framework for The Hundred (Women's Cricket)

Understanding Batting, Bowling, and MVP Impact

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Chapter 1

Introduction

1.1 Motivation Behind This Project

Cricket analytics has traditionally relied on simple statistics such as runs scored, batting averages, wickets taken, and economy rates. While these numbers provide a quick summary of performance, they fail to explain *how* and *when* the performance occurred.

In modern short-format cricket, particularly in tournaments such as **The Hundred**, matches are often decided by a few crucial deliveries. In such formats, a player's impact cannot be judged solely by totals; instead, the match context, pressure situation, and timing of contributions become critically important.

This project was motivated by the need to answer a simple but powerful question:

> *Can we evaluate cricket performance in a way that reflects real match impact rather than just raw numbers?*

1.2 Why The Hundred Requires Context-Aware Evaluation

The Hundred is a unique format with only 100 balls per innings. This leads to:

- Rapidly changing match situations
- High pressure on every delivery
- Limited opportunities for recovery

As a result, traditional long-format metrics become unreliable indicators of performance. This project adopts a **ball-by-ball analytical approach** to capture the true nature of impact in this format.

1.3 Project Objectives

The objectives of this project are:

1. To evaluate batting and bowling performances at match level
2. To incorporate pressure, phase difficulty, and responsibility
3. To extend match ratings to tournament-level rankings
4. To identify best batters, best bowlers, and tournament MVPs
5. To present results in a clear, explainable, and visual manner

Chapter 2

Dataset and Data Preparation

2.1 Understanding the Dataset

The dataset used in this project consists of ball-by-ball CSV files from multiple seasons of The Hundred (Women's Tournament). Each CSV file represents one match and contains details of every delivery bowled.

Ball-by-ball data enables precise tracking of:

- Scoring patterns
- Pressure situations
- Wicket events
- Match progression

2.2 Why Legal Ball Identification Is Necessary

In The Hundred format:

- Wides and no-balls do not count as legal deliveries
- Only legal deliveries contribute to run rate calculations

Therefore, identifying legal balls is essential to avoid distorted pressure calculations.

$$LegalBall = \begin{cases} 1, & \text{if the delivery is legal} \\ 0, & \text{if the delivery is a wide or no-ball} \end{cases}$$

Chapter 3

Batting Performance Analysis (Single Match)

3.1 What Defines Batting Impact

A batter's contribution cannot be judged only by runs scored. True batting impact depends on:

- The phase of the match
- Boundary contribution
- Dot balls consumed
- Pressure under which runs were scored

This project introduces multiple metrics to capture each of these aspects.

3.2 Phase-Based Weighting of Runs

Runs scored in different phases do not have equal importance. Runs in the death phase often have a greater effect on the match outcome.

$$PhaseWeightedRuns = \sum(Runs \times PhaseWeight)$$

3.3 Boundary Impact Score (BIS)

Boundaries have a disproportionate influence on momentum and required run rate.

$$BIS = 1.5 \times Fours + 2.0 \times Sixes$$

3.4 Dot Ball Cost (DBC)

Dot balls increase pressure, especially in short formats.

$$DBC = \sum (DotBalls \times PhasePenalty)$$

3.5 Unified Pressure Index (UPI)

UPI measures how demanding the match situation was during a batter's innings.

3.6 Innings Diversification and Context Awareness

Batting first and batting second present fundamentally different challenges.

3.6.1 Batting Second (Chasing)

$$UPI_{chasing} = \frac{RequiredRunRate}{MatchRunRate}$$

3.6.2 Batting First (Setting a Target)

$$UPI_{setting} = \frac{MatchRunRate - CurrentRunRate}{MatchRunRate}$$

3.7 Final Batting Match Rating

$$BattingRating = (PDB + 8 \times BIS - 6 \times DBC - DP) \times UPI$$

striker	Batting_Match_Rating	RIS	BIS	DBC	PDB	DP
AE Jones	56.000000	33	9.5	4.0	33.0	15
ND Dattani	45.395696	34	8.0	4.8	34.0	15
E Jones	31.360000	47	13.5	16.8	47.0	15
DR Gibson	25.600000	8	3.0	0.0	8.0	0
DB Sharma	12.320000	28	4.5	5.6	28.0	15
GA Elwiss	12.160000	8	1.5	0.8	8.0	0
HC Knight	10.765373	21	4.5	4.8	21.0	15

Figure 3.1: Batting Impact Rating – Match Leaderboard

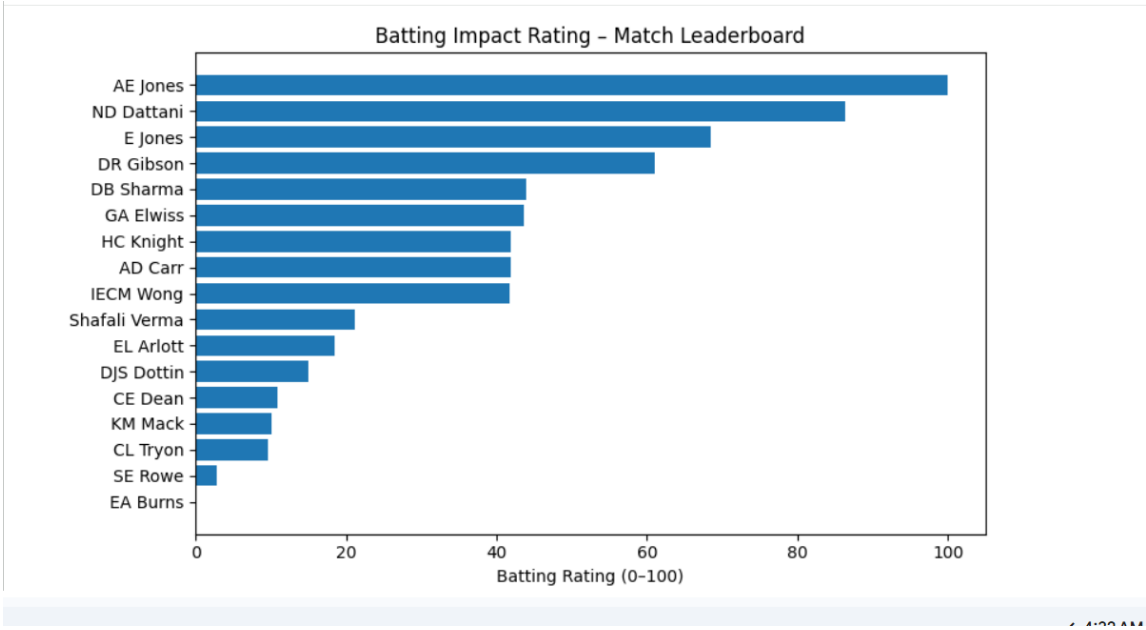

Figure 3.2: Batting Rating Component Breakdown

3.8 Implementation of Batting Metrics (Single Match)

The complete implementation of the batting metrics discussed in this chapter was carried out using Python in a Google Colab environment. This notebook demonstrates how raw ball-by-ball data is transformed into context-aware batting impact scores using the metrics defined above.

Batting Metrics Colab Notebook:

https://colab.research.google.com/drive/1t8qYI6j9F7rt54ocUJGjS1pblw6M203q

Chapter 4

Bowling Performance Analysis (Single Match)

4.1 Understanding Bowling Impact

A bowler's effectiveness goes beyond wickets.

4.2 Wicket Impact Score (WIS)

$$WIS = \sum(Wickets \times PhaseWeight)$$

4.3 Run Pressure Index (RPI)

$$RPI = (MatchRunRate - BowlerRunRate) \times BallsBowled$$

4.4 Final Bowling Match Rating

$$BowlingRating = (20WIS + 15RPI + 10DBP + 10BCI + 5DODB) \times UPI$$

```
result
```

bowler	Match_Bowling_Rating	WIS	RPI	DBP	BCI	DODB
EA Burns	335.959184	2.4	10.530612	12.0	1.000000	0.0
FR Davies	241.959184	1.2	7.530612	9.6	0.900000	0.0
CE Dean	166.802721	1.2	4.897959	6.0	0.933333	0.0
KL Gordon	163.459184	2.4	1.530612	8.4	0.850000	0.0
DJS Dottin	152.959184	2.4	-1.469388	12.0	0.700000	0.0
DB Sharma	136.459184	1.2	0.530612	9.6	0.850000	0.0
HC Knight	98.979592	1.2	1.265306	4.8	0.800000	0.0
IECM Wong	24.459184	1.2	-4.469388	6.0	0.750000	0.0
AH Maqsood	14.489796	0.0	-0.367347	1.2	0.800000	0.0
GA Elwiss	6.117347	0.0	-1.775510	2.4	0.875000	0.0
EL Arlott	-1.530612	2.4	-10.102041	9.6	0.600000	0.0
DR Gibson	-17.020408	0.0	-4.734694	4.8	0.600000	0.0
ND Dattani	-20.510204	0.0	-3.367347	2.4	0.600000	0.0

Figure 4.1: Bowling Impact Rating – Match Leaderboard

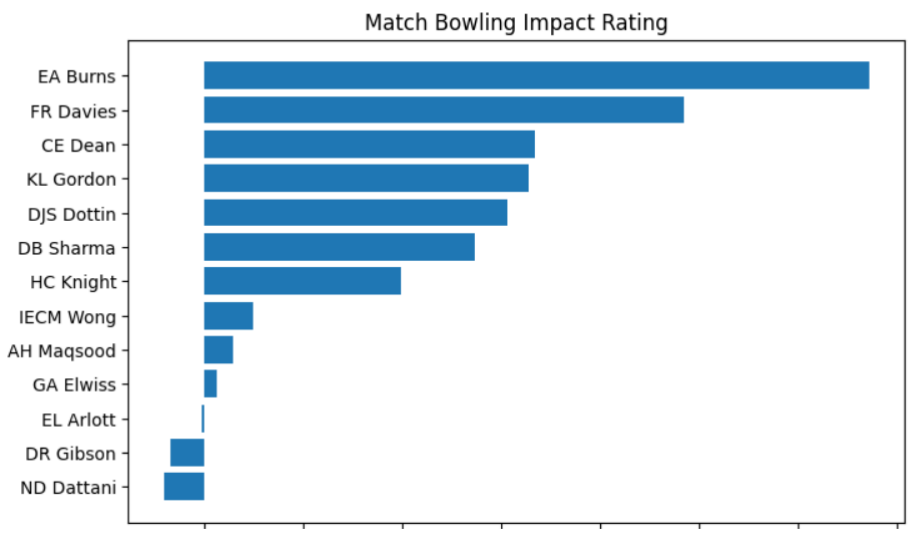

Figure 4.2: Bowling Rating Component Breakdown

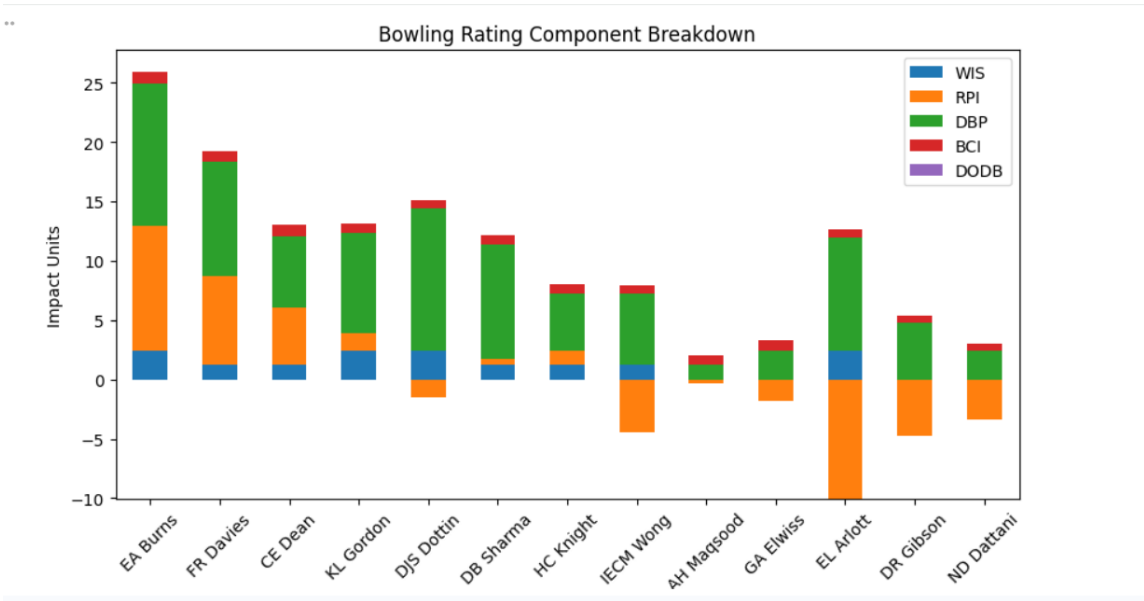

Figure 4.3: Advanced Bowling Pressure and Control Analysis

4.5 Implementation of Bowling Metrics (Single Match)

The bowling impact framework presented in this chapter was implemented in a dedicated Google Colab notebook, ensuring modular and reproducible analysis of single-match data.

Bowling Metrics Colab Notebook:

`https://colab.research.google.com/drive/1KMUQOCdkFo0x1YpexnpgD5wxcquxoIk5`

Chapter 5

Tournament-Level Evaluation

5.1 Why Tournament Aggregation Is Required

Tournament-level evaluation rewards consistency and sustained impact.

5.2 Consistency Factor

$$Consistency = 1 - \frac{StdRating}{AvgRating}$$

5.3 Experience Factor

$$Experience = \log(1 + MatchesPlayed)$$

5.4 Tournament Impact Score

$$TournamentScore = AvgRating \times Consistency \times Experience$$

	season	striker	Batting_Rating
127	2021	NR Sciver-Brunt	97.200000
248	2022	RA Fackrell	69.200000
347	2023	SIR Dunkley	68.157143
415	2024	L Filer	77.300000
492	2025	G Wareham	91.000000

Figure 5.1: Top Batters – Tournament Impact Rating

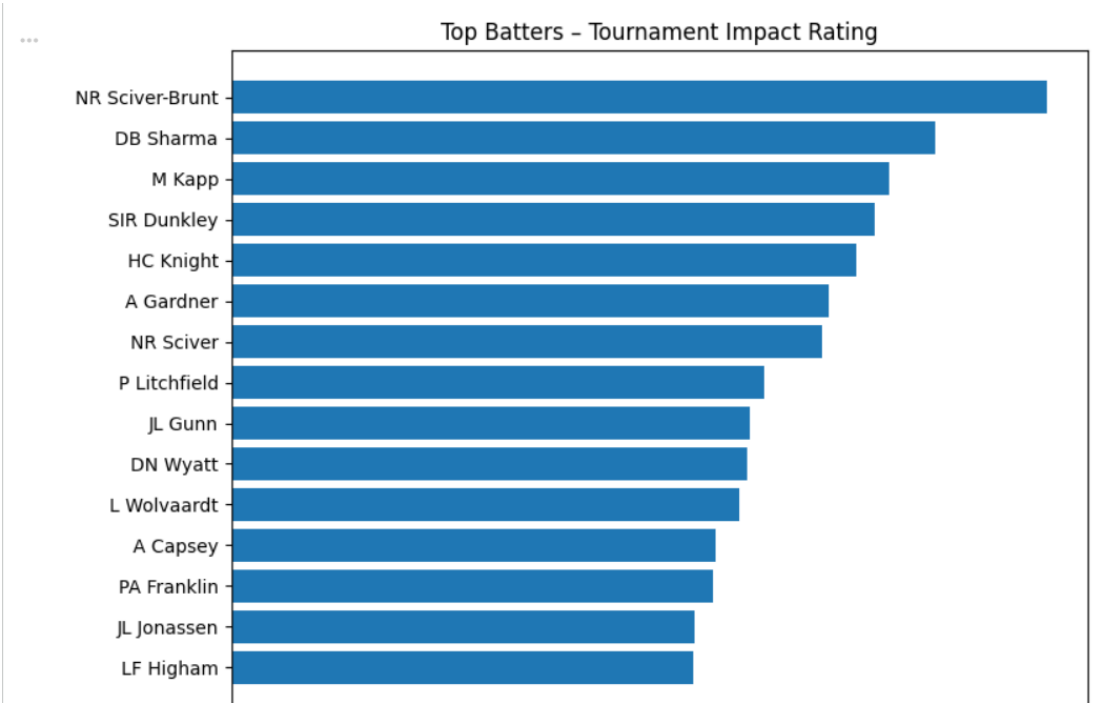

Figure 5.2: Batting Consistency vs Tournament Impact

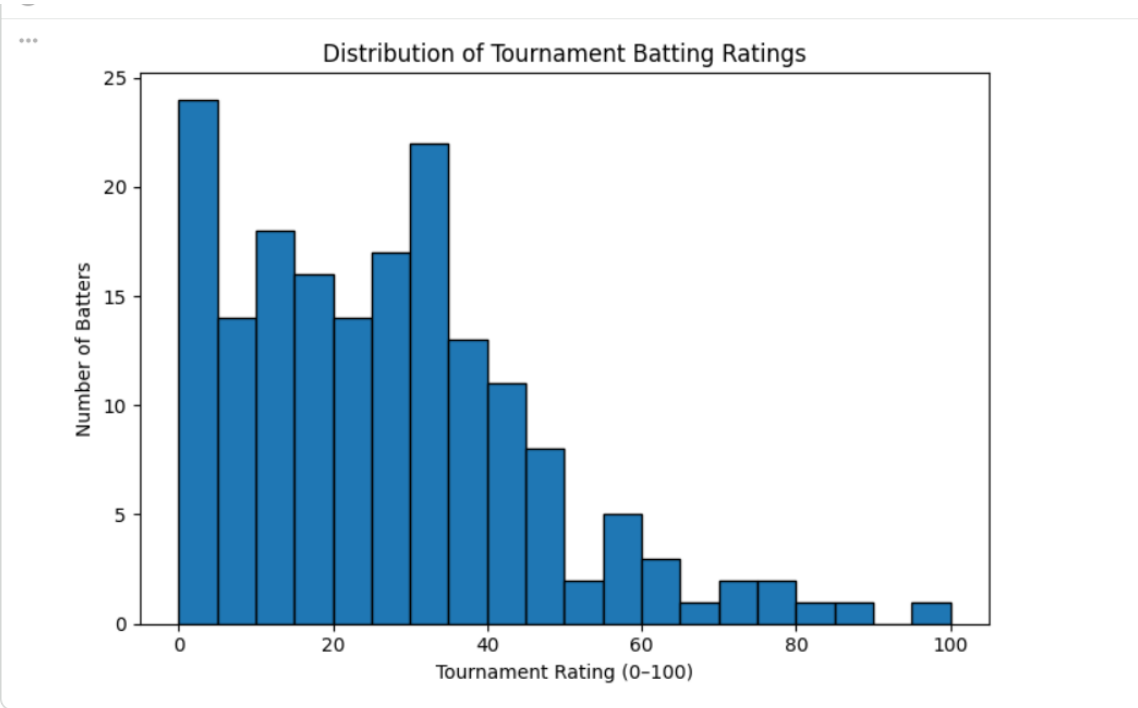

Figure 5.3: Distribution of Tournament Batting Ratings

bowler	Matches	Avg_Rating	Std_Rating	Avg_UPI	Peak_Rating	Consistency	Experience	Tournament_Bowling_Score	Tournament_Rating_0
l Kapp	34	71.885294	31.716224	0.894638	100.00	0.558794	3.555348	142.815025	100.00
A therland	24	63.320833	24.356358	0.862017	97.21	0.615350	3.218876	125.421809	87.82
A rubsole	27	61.888889	24.686781	0.878593	92.80	0.601111	3.332205	123.965030	86.80
S clestone	34	63.370588	31.080126	0.882473	100.00	0.509550	3.555348	114.803831	80.38
Ismail	34	59.270588	27.046341	0.866949	96.22	0.543680	3.555348	114.568415	80.22
N Smith	38	61.773684	30.970428	0.890764	100.00	0.498647	3.663562	112.849629	79.01
. Cross	41	57.656098	28.201073	0.878181	89.30	0.510874	3.737670	110.093150	77.08
J Brett	2	100.000000	0.000000	1.001181	100.00	1.000000	1.098612	109.861229	76.92
.K Bell	42	61.066667	33.798475	0.899279	100.00	0.446532	3.761200	102.561126	71.81

Figure 5.4: Top Bowlers – Tournament Impact Rating

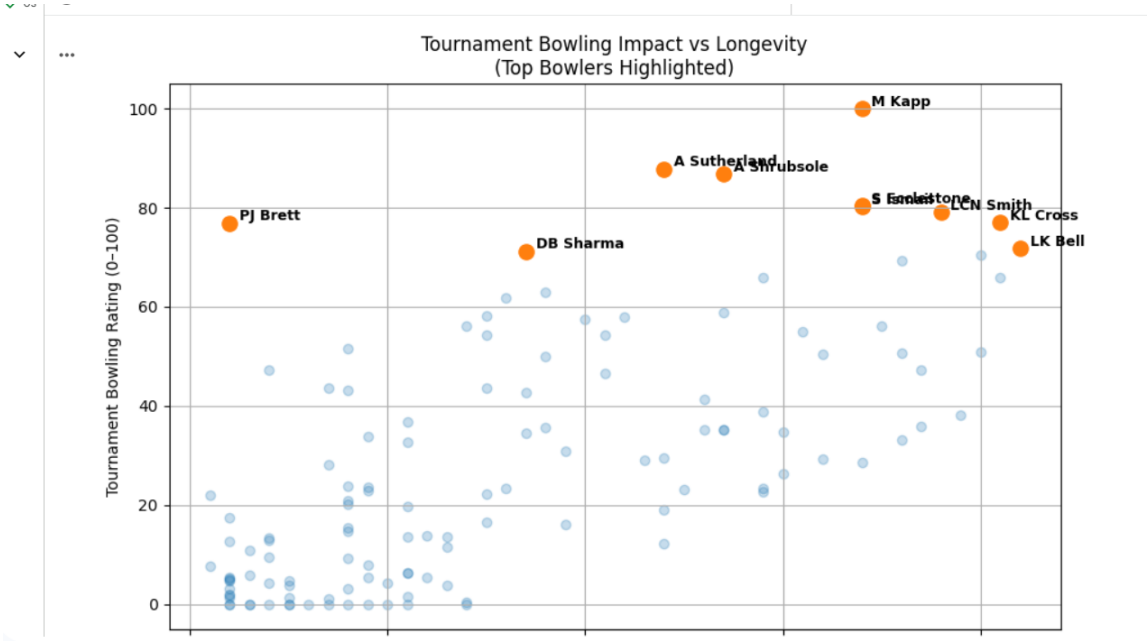

Figure 5.5: Bowling Consistency vs Tournament Impact

5.5 Implementation of Tournament-Level Analytics

Tournament-level aggregation and MVP computation were implemented using a dedicated Google Colab notebook that executes match-level pipelines across all available match files.

Tournament Analytics Colab Notebook:

`https://colab.research.google.com/drive/1P7TThEES9Akx5IqfIXIIn9Tl2ZhjUEAW`

Chapter 6

MVP Analysis

6.1 Rationale Behind MVP Index

$$MVP = 0.6 \times BattingRating + 0.4 \times BowlingRating$$

	Matches_bat	Avg_Rating_bat	Std_Rating_bat	Avg_UPI_bat	Peak_Rating_bat	Consistency_bat	Experience_bat	Tournament_Ba
M Kapp	34.0	52.402941	31.993517	0.827585	100.00	0.389471	3.555348	
DB Sharma	13.0	56.284615	26.850507	0.823342	85.36	0.522951	2.639057	
NR Sciver-Brunt	24.0	65.837500	37.864286	0.818370	100.00	0.424883	3.218876	
A Sutherland	24.0	47.066667	32.715661	0.813549	97.33	0.304908	3.218876	
A Gardner	17.0	53.629412	30.836621	0.811981	100.00	0.425005	2.890372	
S Ecclestone	28.0	36.953571	24.781960	0.813727	73.18	0.329376	3.367296	
LCN Smith	13.0	30.053846	15.945617	0.814268	52.02	0.469432	2.639057	

Figure 6.1: Tournament MVP Space – Batting vs Bowling Contribution

Chapter 7

Conclusion and Future Scope

7.1 Key Contributions of This Work

This project demonstrates an explainable, context-aware framework for cricket performance evaluation in The Hundred format.

7.2 Future Extensions

Future work may include:

- Machine learning–based performance prediction
- Player role classification
- Team strategy optimisation